7

Time and the Quantum

INSIGHTS INTO TIME'S NATURE FROM THE QUANTUM REALM

When we think about something like time, something we are within, something that is fully integrated into our day-to-day existence, something that is so pervasive, it is impossible to excise—even momentarily—from common language, our reasoning is shaped by the preponderance of our experiences. These day-to-day experiences are classical experiences; with a high degree of accuracy, they conform to the laws of physics set down by Newton more than three centuries ago. But of all the discoveries in physics during the last hundred years, quantum mechanics is far and away the most startling, since it undermines the whole conceptual schema of classical physics.

So it is worthwhile to expand upon our classical experiences by considering some experiments that reveal eyebrow-raising features of how quantum processes unfold in time. In this broadened context, we will then continue the discussion of the last chapter and ask whether there is a temporal arrow in the quantum mechanical description of nature. We will come to an answer, but one that is still controversial, even among physicists. And once again it will take us back to the origin of the universe.

The Past According to the Quantum

Probability played a central role in the last chapter, but as I stressed there a couple of times, it arose only because of its practical convenience and the utility of the information it provides. Following the exact motion of the 10^{24} H_2O molecules in a glass of water is well beyond our computational capacity, and even if it were possible, what would we do with the resulting mountain of data? To determine from a list of 10^{24}